

SSRA: Scale-Shared Recursive Attention — Design and Complexity Analysis

Technical note (stage 1 of 2).

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Versioning: This note fixes the design and its complexity analysis *before* any training run. A stage-2 paper with experimental results will follow the evaluation plan in §8 and will cite this note as the prior version.

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Abstract

We describe SSRA (Scale-Shared Recursive Attention), a causal language-modeling block in which a single weight-shared rule — one softmax attention block Attn_θ and one learned pooling operator Pool_ϕ per layer — generates the entire scale hierarchy of the sequence. The same (θ, ϕ) is applied at every level of a binary tree built over the sequence, and the same θ performs the token-level read-out: a token is treated as a level-0 node. Two further properties complete the mechanism: learned, discrete-slot compression at every node (m summary vectors per node, via latent-query attention or hard top- m selection), and fully bidirectional attention inside nodes that remains legal in a causal LM because causality is enforced structurally — a node becomes consumable only after its span is complete — rather than by masking. Training cost is $\Theta(N \cdot (w + m \cdot \log N) \cdot d)$ per layer and the autoregressive decoding state is $O(m \cdot d \cdot \log N)$ per layer; this is the same complexity class as Log-Linear Attention, and we explicitly do not claim a better class. The contribution of this note is the design itself, its causality proof, its derived complexity, and a pre-registered falsification plan: two ablation axes (cross-scale weight sharing on/off; pooling operator family) and a multi-needle retrieval suite decide, at matched compute, whether the mechanism earns its place. Either outcome — positive or negative — answers the underlying hypotheses and is intended for publication.

1. Motivation

Natural language exhibits self-similar, long-range-dependent statistics: Alabdulmohsin et al. estimate a Hurst parameter of $H \approx 0.7$ for language and connect this fractal structure to next-token prediction [1]. Standard Transformers do not encode this property architecturally: every layer attends over a flat token sequence, and hierarchical variants in the literature typically introduce *per-level* parameters over a small, fixed number of levels.

SSRA tests a sharper inductive bias: **if language is self-similar, one rule should suffice at every scale.** Concretely, a single attention block and a single learned compressor per layer process pairs of tokens, pairs of summaries of pairs, and so on up a binary tree — and the very same attention block performs the final token-level read-out. The hypotheses this design operationalizes are:

- **H1 (scale-shared rule):** cross-scale weight sharing is an inductive bias matched to the self-similarity of language [1], yielding better length extrapolation at equal parameter budget than a flat baseline.
- **H2 (learned node compression):** m -slot learned compression at tree nodes preserves retrieval ability (measured by needle-in-a-haystack tests) at $\Theta(N \cdot (w + m \cdot \log N) \cdot d)$ training cost.

Both hypotheses are falsifiable by the pre-registered ablations in §8; the failure modes are spelled out in §6.

2. Mechanism overview

SSRA replaces the attention sublayer of a standard pre-norm Transformer block:

```
x ← x + SSRA_mix_i(LN1(x))      # this note
x ← x + FFN_i(LN2(x))          # standard MLP; not used inside tree nodes
```

Each layer i owns one attention parameter set θ_i (W_Q, W_K, W_V, W_O ; h heads), one pooling parameter set ϕ_i , and a learned level-embedding table e_ℓ . All three are **shared across all tree levels, all nodes, and the read-out within that layer**; nothing is shared across layers in this version. Each layer builds its own tree over its own current activations; summaries are per-layer activations, not persistent state.

Notation: sequence length N ; model width d ; tree level ℓ (tokens are level-0 nodes); node u at level ℓ spans 2^ℓ consecutive tokens; summary budget $m = 16$ (default); per-level summary count $s_\ell = \min(2^\ell, m)$; local read-out window $w = 64$ (default); tree arity $k = 2$ (default).

2.1 Up-pass (tree construction)

```
leaf u = token t:                S_u = z_t                # s_0 = 1
internal node u (children c1, c2, level ℓ ≥ 1):
  X_u = concat(S_c1, S_c2)        # ≤ 2m input slots
  X̃_u = X_u + e_ℓ                 # level coordinate as input
  H_u = X̃_u + Attn_θ(LN_node(X̃_u)) # pre-norm residual;
                                   # bidirectional over slots;
                                   # relative slot-RoPE (positions 1..n_in)
                                   # identity while s_ℓ = n_in
  S_u = Pool_φ(H_u, s_ℓ)
```

Pre-norm residual paths and a LayerNorm in every node are hard requirements: without an identity path, log N successive rewrites destroy both verbatim signal transport and gradient flow. With fixed $m = 2^a$, levels $\ell \leq a$ are lossless (Pool = identity); the first lossy compression occurs at level $a+1$ ($32 \rightarrow 16$ slots for $m = 16$). For ragged N, a node is materialized iff its span lies inside $[1, N]$; no padding, no partial nodes. All nodes of a level are processed as one batched tensor (level-wise batching); recursion in the host language is prohibited in the training path.

2.2 Pooling operators Pool_φ

Three operators sit behind one interface (one operator per run, same operator at every lossy level):

operator	role	definition (sketch)	anchor
P1 latent-query attention pool	default	s_ℓ learned latent queries cross-attend to the node's slots, reusing the layer's θ projections; ϕ adds only the queries and one LN	PMA / Set Transformer [16], Perceiver [17]
P2 strided pairwise merge	control	learned linear merge of adjacent slot pairs ($2 \rightarrow 1$)	Hourglass-style shortening [9]; token merging [19]
P3 learned top- $(s_\ell-1)$ selection + context residual	challenger	linear scorer; top slots copied verbatim (exact content pass-through), one extra slot is a temperature-softmax summary of the non-selected slots; straight-through or Gumbel-top-k gradients in training only; inference is hard and deterministic	natively trainable selection [6]

P1's latent queries are shared across **all** nodes and **all** scales of the tree — the maximal form of the scale-sharing bet, and (to our knowledge) a new use of PMA-style pooling. P2 isolates the question of whether selection must live in the pooling operator at all, or whether the shared Attn_θ can do the selection on its own. P3 trades smoothness for verbatim transport and is subject to a pre-registered stability gate (§8). A hybrid (k_{sel} verbatim slots + remaining latent-query slots) is a prepared fallback challenger.

2.3 Positional scheme: shared geometry, coordinate as input

location	treatment
inside a node	relative slot-RoPE over slot indices $1..n_{\text{in}}$ — <i>identical geometry at every level</i> , which is exactly what makes the rule scale-transportable (the H1 bet); left/right child encoded by slot order
node input	learned level embedding e_ℓ added to every slot (initialized at zero; ablation OFF tests pure scale equivariance)
read-out, window keys	RoPE at absolute token positions (relative offsets $\leq w$ by construction)
read-out, summary keys	no rotary phase (NoPE) + e_ℓ added to keys only — the coordinate modulates addressing, not content

In the Fenwick decomposition (§2.5) the level of a block grows with its distance from the query, so e_ℓ doubles as a log-scaled distance code: recent context is addressed finely, old context coarsely. Mixing rotated token keys with special non-rotated keys in a single softmax follows the attention-sink / memory-token precedent [20], with one honest caveat: [20] assigns its persistent keys positions *within the cache*, i.e. the directly precedented treatment is a fixed virtual position rather than pure NoPE. Pure-NoPE summary keys are therefore a hypothesis checked in the first implementation milestone, with a fixed-virtual-position fallback specified in the design.

2.4 Causality without masks inside nodes

Three mechanisms, no causal mask inside nodes:

1. **Local window:** the read-out query at position t attends to tokens $[\max(1, t-w), t]$ only.
2. **Structural gating:** a node u is consumable by the read-out only once $\text{span}(u) \subseteq [1, t-w-1]$, and by its parent only when its sibling's span is also complete. Causality is an availability property of summaries, not a mask.
3. **Bidirectional attention inside nodes is legal.** Proof sketch: a summary S_u can influence the logit at position t only through an ancestor-or-self a that the read-out at t consumes, which requires $\text{span}(u) \subseteq \text{span}(a) \subseteq [1, t-w-1]$; hence every token inside u precedes t , and no information flows from positions $\geq t$ into the prediction at t . ■ The enumeration of paths is exhaustive: the only cross-node operation in the up-pass is parent composition — attention, normalization, and pooling all act within a single node — so the dataflow graph has no edges besides child-to-parent composition and the structurally gated read-out.

This buys an expressivity asymmetry over a flat causal Transformer: *within* a completed span, attention is unrestricted in both directions, because every consumer of that span sits strictly after it. A lesson encoded from an earlier prototype of this project: a decreasing training loss is *not* evidence of causal correctness (target leakage produces the same curve); only explicit tests are. Two are pre-registered: a **shift test** (perturbing token t must not change any logit at positions $< t$) and a **completion test** (incremental decoding must reproduce the full batched forward bit-for-bit within tolerance at every position).

2.5 Read-out (Fenwick prefix decomposition)

For query position t with prefix budget $p = t - w - 1$, the prefix $[1, p]$ is decomposed by the standard binary-indexed-tree (Fenwick) rule into $\leq \lceil \log_2 p \rceil + 1$ disjoint blocks, each of which is exactly a tree node (block ends at a multiple of its size; always materialized). The key/value set for position t is

$$K_t = \{ z_j : j \in [\max(1, t-w), t] \} \cup \{ \text{rows of } S_u : u \in \text{Fenwick}(t-w-1) \}$$

$$y_t = \text{Attn}_\theta(q = z_t, kv = K_t) \quad \# \text{ ONE softmax over heterogeneous keys}$$

so $|K_t| \leq (w+1) + m \cdot (\lceil \log_2 t \rceil + 1)$, and window + Fenwick blocks cover $[1, t]$ disjointly and completely. The read-out **reuses θ** — the same attention block as in the nodes; a token is a level-0 node. Per-block capacity is m vectors *per block*, i.e. the macro context available to a query scales as $m \cdot \log t$ vectors, not a single m -vector bottleneck.

The Fenwick prefix structure itself is taken from Log-Linear Attention [2]; we claim no novelty for it (§6).

2.6 Autoregressive decoding

Per layer, the decoder maintains a ring buffer of the last $w+1$ token states and a node store governed by an explicit retention rule: at time t , retain node u iff $u \in \text{Frontier}(t) = \text{fenwick_blocks}(t)$ (needed to build future parents) **or** $u \in \text{Fenwick}(t-w-1)$ (needed by the read-out now). The naive rule "free children when the parent forms" is incorrect: the read-out frontier lags the tree frontier by $w+1$ positions, so children remain consumable for $w+1$ steps after their parent completes. Both sets have $\leq \lceil \log_2 t \rceil + 1$ members, giving per-layer state $\leq (w+1) \cdot d + 2 \cdot m \cdot d \cdot (\lceil \log_2 N \rceil + 1) = O(m \cdot d \cdot \log N)$. Amortized, one internal node completes per generated token; the worst case at $t = 2^k$ is a chain of k sequential completions — an $O(\log N)$ latency spike, accepted for the proof of concept and documented as a limitation.

3. Complexity (derived; per layer, $\times L$ for the model)

Score-operation accounting, binary tree, pass-through schedule $s_\ell = \min(2^\ell, m)$:

quantity	bound	derivation sketch
up-pass, training	$O(N \cdot m \cdot d)$	below the lossy threshold $\sum_\ell c \cdot N \cdot 2^\ell \cdot d < 2cNmd$; above it $\sim N/m$ nodes at $4c \cdot m^2 \cdot d$ each $\Rightarrow < 4cNmd$
read-out, training	$O(N \cdot (w + m \cdot \log N) \cdot d)$	position t sees $\leq (w+1) + m(\lceil \log_2 t \rceil + 1)$ keys; sum over t
training total	$O(N \cdot (w + m \cdot \log N) \cdot d)$	read-out dominates; with w, m, d constant this is $O(N \log N)$ — the same class as Log-Linear Attention [2], not better
activation memory (tree)	$O(N \cdot d \cdot \log m)$	$\approx 5\times$ flat token activations at $m = 16$; score matrices are $N \cdot (w + m \cdot \log N)$ vs N^2 flat
decode state	$O(m \cdot d \cdot \log N)$	retention rule of §2.6, explicit constant 2 on the log term
decode compute / token	$O((w + m \cdot \log N + m^2) \cdot d)$ amortized	read-out + amortized one node completion
decode worst-case latency	$O(\log N)$ sequential completions at $t = 2^k$	documented limitation

Honest accounting. The QKVO projections add $O(N \cdot d^2)$ terms. These are class-neutral, but at $d \approx 512$ and $N = 8k$ they are of the same order as the score operations. Consequently, any wall-clock advantage over a flat Transformer is an *empirical* question decided by measured throughput curves (§8), not by the asymptotics above. We do not claim a wall-clock advantage in this note. For calibration: a flat Transformer trains at $O(N^2 \cdot d)$ with $O(N \cdot d)$ decoding state; H-Transformer-1D reaches linear time and memory by hierarchical low-rank approximation [3]; Log-Linear Attention trains at $O(T \log T)$ with $O(\log T)$ state [2]; prefix-scannable models reach $O(\log N)$ state with softmax-like aggregation operators [23]. SSRA's claim is not a better complexity class than [2] or [23] — it is a different mechanism inside the same class, and §8 specifies the measurements that decide whether that mechanism is worth having.

4. Parameter inventory

Per layer, SSRA's overhead relative to a flat pre-norm Transformer at equal d, h, L is the level-embedding table, one node LayerNorm, and ϕ . At $d = 512, m = 16, N_{\text{max}} = 32k$ this is $\approx 26k$ parameters per layer — well under 1 % of a layer. P1's ϕ is $m \cdot d + 2d$ (latent queries + a LayerNorm; the attention projections are reused from θ); P2's is $2d^2 + d$; P3's is $d + 1$. Parameter-matched comparisons with the flat baseline are therefore nearly free.

5. What this design deliberately is not (non-goals)

No top-down / coarse-to-fine output pass (a candidate follow-up, see §7). No content-based or learned segment boundaries — the hierarchy is strictly positional; learned boundaries are an orthogonal axis explored by dynamic token pooling [25] and H-Net [13]. No claim of a better complexity class than [2]. No KV-cache tricks, quantization, or kernel engineering beyond what the throughput measurement needs. No inference-time stochasticity of any kind.

6. Novelty thesis and delimitation

Claim. SSRA is a causal language-modeling block in which a single weight-shared rule — one softmax attention block Attn_θ and one learned pooling operator Pool_ϕ per layer — generates the entire scale hierarchy of the sequence. The same (θ, ϕ) applies at every level of a binary tree over the sequence, and the same θ performs the token-level read-out: a token is a level-0 node. Two properties complete the mechanism: (i) learned, discrete-slot compression at every node (m summary vectors via latent-query attention or hard top- m selection), and (ii) fully bidirectional attention inside nodes that remains legal in a causal LM because causality is enforced structurally — a node becomes consumable only after its span is complete — rather than by masking.

Delimitation. The prefix decomposition used by the read-out is the standard Fenwick (binary indexed tree) structure shared with Log-Linear Attention [2]; we claim no novelty for it. Relative to [2], the mechanism differs in all three remaining components: softmax attention vs. linear-attention kernels; learned discrete-slot pooling vs. structural matrix state with λ -decay; and cross-scale parameter sharing that includes the read-out. Relative to hierarchical and multiresolution models [3, 8, 9], SSRA shares one rule across all levels instead of per-level parameters in fixed 2–3-level stacks; notably, Fast Multipole Attention learns its group-compression weights but learns them *per level* and aggregates linearly rather than with a shared attention block [4],

and MRA attention uses a fixed wavelet family [5]. Relative to weight sharing in depth [10, 11], sharing here is across sequence scale. PMA/Perceiver-style pooling [16, 17] anchors the P1 operator; its use as the node compressor of a causal scale hierarchy with queries shared across all nodes and scales is, to our knowledge, new. The nearest combinations found in a verification pass and an arXiv novelty scan dated 2026-06-11 are delimited explicitly: GPST [21] shares a learned composition function over an induced *syntactic* tree inside a generative LM, but composes constituents to single vectors, keeps the composition model separate from the generative model, and makes no long-context or retrieval claim; MANO [22] shares convolution kernels and attention weights across scales, but for encoder-style vision/physics grids without causality, slot summaries, or a prefix read-out; prefix-scannable models [23] obtain softmax-like operators with $O(\log N)$ decoding state via prefix scans over chunk encodings, without cross-scale rule sharing or learned per-node slot compression; the Hierarchical Kernel Transformer [24] computes independent score matrices per resolution from trainable causal downsampling and fuses them, again without a shared rule or node summaries. No work among the surveyed set [1–25] simultaneously combines: cross-scale weight sharing $\times$ learned m-slot node compression $\times$ causal LM over text. (SSRA is also unrelated, beyond the name, to Fractal Generative Models for images [15].)

Falsifiability. The thesis is decided by two pre-registered ablation axes at matched compute: if cross-scale sharing does not help (ablation a, §8), SSRA degenerates into a slower relative of H-Transformer-1D [3]; if learned node compression loses retrievable detail (multi-needle suite, prediction P-B, §8), SSRA loses to Log-Linear Attention [2] on recall. Either outcome answers H1–H2 and is publishable.

7. Deferred design branch: top-down output path

A second output-path variant — a full top-down cascade in which coarse states condition finer ones — was designed and rejected for this version: at representative hyperparameters it is substantially more expensive to train, its macro state is stale by up to $2^{\wedge}l$ tokens at level l , and it routes the entire macro context through m vectors per node, a structural bottleneck on exactly the retrieval axis where the empirical contest with [2] is decided. It is retained as a named follow-up branch (SSRA-TD), motivated by a separate hypothesis (top-down conditioning improves long-generation consistency) that this proof of concept does not test.

8. Pre-registered evaluation plan

Verification before any training conclusion (milestone M1). Shift test (causality), completion test (incremental $\equiv$ batched forward at every position; certifies the Fenwick logic and the decode retention rule simultaneously), gradient-flow check across all parameter groups, and throughput/VRAM curves vs $N \in \{1k..8k+\}$ against a flat baseline at equal d, h, L . Acceptance proposal for the throughput curve: log-log slope ≤ 1.5 over the range and strictly below the flat baseline's slope. A stability micro-gate for P3: on a $\sim 10M$ -parameter smoke run, P3 must reach a loss within 5 % (relative) of P1 without divergence after standard stabilization (load-balance auxiliary loss, temperature annealing); failure demotes P3 to an appendix and promotes the $P1 \times P3$ hybrid to challenger.

Training (M2). Two scales ($\approx 25M$ for tuning, $\approx 85M$ for the main comparison) on a public web-text sample, token budget $\approx 20\times$ parameters, training context 1k–8k. Gate: stable training and short-context perplexity within ± 5 % of the compute-matched flat baseline.

Evaluation (M3). Baselines: (a) compute-matched flat Transformer; (b) Log-Linear Attention / Gated DeltaNet from a public implementation [2]; (c) a MEGABYTE-style two-level model [8]. Metrics: perplexity vs context length (1k $\rightarrow$ 32k, beyond training length), per-position loss, needle-in-a-haystack and a **multi-needle suite** ($k \in \{1, 2, 4, 8, 16\}$ needles, distances from 128 to full context, exact-match recall), throughput and memory in training and decoding (including the worst-case completion spike), and pooling-collapse diagnostics (entropy of latent-query attention maps, per-query participation).

Pre-registered predictions.

- **P-A:** with P2 (control), needle recall decays roughly with the number of compression levels between needle and query; $P1/P3$ stay approximately flat up to a capacity ceiling.
- **P-B (sharpest discriminator):** k needles inside one subtree produce a recall cliff near $k \approx m$ for the learned operators ($P3$ near $m-1$; $P1$ earlier if query collapse occurs) and near $k \approx 1$ for $P2$.
- **P-C:** latent-query collapse, if it occurs, is visible in training-time entropy/participation diagnostics before it is visible in task metrics.

Ablation axes (registered). (a) cross-scale weight sharing on/off (per-level θ copies) — the core H1 test; (b) $\text{Pool}_{\phi} \in \{P1, P2, P3, \text{hybrid}\}$; (c) m fixed vs linearly growing schedule; (d) $w \in \{32, 64, 128\}$; (e) read-out parameters shared vs separate; (f) level embedding on/off; (g) arity $k \in \{2, 4\}$ and (h) the induced depth change. The $k = 2$ default is chosen deliberately: it matches the Fenwick skeleton of [2] exactly, so the comparison isolates the mechanism rather than the tree structure.

Outcome policy. Success is defined as a clear, publishable answer to H1–H2 at matched compute — positive or negative. A measurable long-context advantage (perplexity tail or needle recall) at $\geq$ equal short-context perplexity yields a positive stage-2 paper; otherwise the stage-2 paper is an analytical negative result locating where and why the mechanism fails.

9. Limitations

Small scale only ($\leq \sim 85M$ parameters); a single text domain; no scaling laws. No experimental results exist at the time of this note — every empirical statement above is a pre-registered plan, not a finding. The top-down branch (SSRA-TD, §7) and content-based hierarchies are untested deferrals. The complexity statements count score operations; QKVO projection terms are of comparable magnitude at the planned widths, so wall-clock conclusions await measurement (§3). Decoding has an $O(\log N)$ worst-case latency spike at power-of-two positions. Activation memory of the tree is $\approx 5\times$ flat token activations at $m = 16$ (score-matrix memory is, however, far below the flat N^2 term at the planned lengths). Pure-NoPE summary keys in the read-out rest on an adjacent rather than exact precedent (§2.3) and carry a specified fallback.

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